

XDJ-X

All-in-one DJ system

Exchange protocol

Service commands

Assembly name:	Firmware version:
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Display module	0.50
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Mixer assy	0.26
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Deck assy	0.17
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Power manager	0.21
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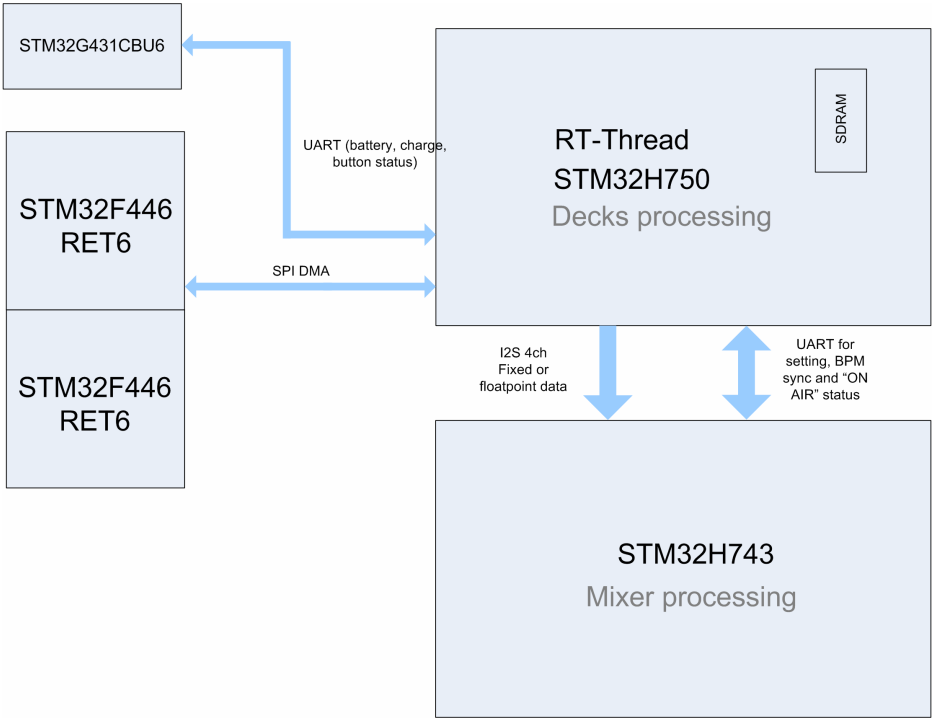
Last modified date:

24.03.2025

Content

1. Exchange protocol	3
2. Service commands	5
3.	5

General connection diagram exchange date:



Display module <=> Power manager

UART transfer 115200 baud.

Power manager => Display module:

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Bytes	0xAX								0xXX							
Bits	1	0	1	0	0	0	X	X	X	X	X	X	X	X	X	X

Byte[0] - status:

A1: no AC

A2: AC charge

A3: full battery

Byte[1] - data:

0...100: charge precentage (when the value changes or on request)

0xFE: button pressed

0xFD: button unpressed

0xFC: shutdown command

Display module => Power manager:

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Bytes	0xA1								0xXX							
Bits	1	0	1	0	0	0	0	1	X	X	X	X	X	X	X	X

Byte[0] = 0xA1

Byte[1]:

0x55: battery percentage request

0xFC: shutdown permission

Display module <=> Deck assy

SPI DMA transfer

Deck assy => Display module:

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
Bytes	PADS								BUTTONS							
Bits	PD7	PD6	PD5	PD4	PD3	PD2	PD1	PD0	RLP	LP8	LP4	VNL	SLIP	REV	CUE	PLAY

	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
Bytes	BUTTONS				ADC_PITCH_MSB				ADC_PITCH_LSB							
Bits	JSNS	TMP	NXT	PREV	X	X	X	X	X	X	X	X	X	X	X	X

	20	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	2E	2F
Bytes	JOG_PULSE_CNT_MSB								JOG_PULSE_CNT_LSB							
Bits	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

	30	31	32	33	34	35	36	37	38	39	3A	3B	3C	3D	3E	3F
Bytes	JOG_SPEED_MSB								JOG_SPEED_LSB							
Bits	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

	40	41	42	43	44	45	46	47
Bytes	CRC							
Bits	X	X	X	X	X	X	X	X

ADC_PITCH 0...4095

JOG_PULSE_CNT 0...65535 16 bits resolution (3600 pulses per 1 round)

JOG_SPEED 0...65535 16 bits resolution (when jog stopped speed = 65535)

CRC = (0byte + 1byte + ... + 7byte) % 256

Display module => Deck assy:

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
Bytes	NUM_P								PADn_RED							
Bits	1	1	0	0	0	X	X	X	R	R	R	R	R	R	R	R

	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
Bytes	PADn_GREEN								PADn_BLUE							
Bits	G	G	G	G	G	G	G	G	B	B	B	B	B	B	B	B

	20	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	2E	2F
Bytes	LEDS/VFD/JOG RING								SLIP_POS							
Bits	RNGr	RNGg	RNGb	0	0	0	LD1	LD0	0	X	X	X	X	X	X	X

	30	31	32	33	34	35	36	37	38	39	3A	3B	3C	3D	3E	3F
Bytes	CUE_POS								PLAY_POS							
Bits	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

	40	41	42	43	44	45	46	47
Bytes	CRC							
Bits	X	X	X	X	X	X	X	X

NUM_P	LD1	LD0
0	CUE	PLAY
1	SLIP	REV
2	LP4	VNL
3	RLP	LP8
4	VNLr	TMP
5	MEMr	SNSr
6	0	0
7	0	0

NUM_P package number and at the same time pad number
 PADn_(RED, GREEN, BLUE) RGB 888 color for PAD[NUM_P]
 RNGr, RNGg, RNGb – RGB bits <<[NUM_P]
 MEMr, SNSr, VNLr – VFD segments
 SLIP_POS 0...84. 85 – SLIP disable
 CUE_POS 0...84. 85 – CUE without cue on jog display
 PLAY_POS 0...134 – jog display position cursor 139 = power on animation 138 = eject animation 137 = load in animation 136 = fill circle on display 135 = empty circle.
CRC = (0byte + 1byte + ... + 7byte) % 256

Display module <=> Mixer assy

UART transfer 115200 baud.

Display module => Mixer assy:

Mixer assy => Display module:
ON AIR